UNIT QUESTIONS

SKILLS INTERPRETATION 8

1

Hydrogen, H_2, and bromine, Br_2, react vigorously to form hydrogen bromide, HBr, according to the following equation:

$$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$$

a Draw a dot-and-cross diagram to show the arrangement of the electrons in a hydrogen bromide molecule, showing electrons in the outer shell only. **(2)**

CHEMISTRY ONLY

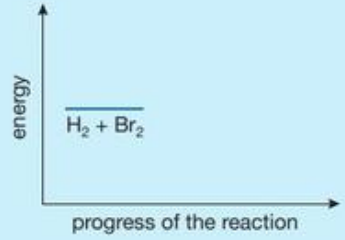
SKILLS PROBLEM SOLVING 10

b Use the bond energies given below to calculate the enthalpy change for the reaction shown in the above equation. **(3)**

Bond	Bond energy/kJ/mol
H–H	436
Br–Br	196
H–Br	368

SKILLS INTERPRETATION 9

c i Complete the reaction profile diagram by showing the products of the reaction. **(1)**



ii Label the diagram to show the enthalpy change, ΔH, and the activation energy of the reaction. **(2)**

(Total 8 marks)

END OF CHEMISTRY ONLY

3

2

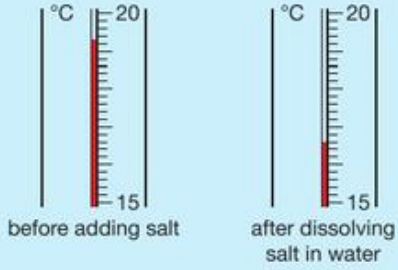
A group of students wanted to investigate the energy changes when salts dissolve in water. The teacher suggested that they should measure the temperature changes that occur when the salts were dissolved.

This is the method they followed:

- Add 100 cm^3 of water to a beaker.
- Record the temperature of the water.
- Weigh 5.00 g of salt and add it to the water in the beaker.
- Stir the mixture with a glass rod vigorously until all the solid has dissolved.
- Record the maximum (or minimum) temperature of the solution.

SKILLS ANALYSIS, PROBLEM SOLVING

a The diagram below shows the readings on the thermometer before and after the student dissolved a salt, potassium chloride, in water.



before adding salt — after dissolving salt in water

i Write down the thermometer readings and calculate the temperature change. **(3)**

Temperature before …….. °C

Temperature after …….. °C

Temperature change …….. °C

SKILLS CRITICAL THINKING 8

ii Which of the following statements is true about the dissolving of potassium chloride in water (choose one answer)? **(1)**

A The process is endothermic and ΔH is positive.

B The process is exothermic and ΔH is positive.

C The process is endothermic and ΔH is negative.

D The process is exothermic and ΔH is negative.

SKILLS PROBLEM SOLVING

b Another student repeated the experiment with a different salt, calcium chloride, using the same method and recorded these results.

Volume of water = 100 cm^3

Starting temperature of water = 15.9 °C

Maximum temperature of solution = 23.2 °C

Mass of salt = 5.00 g

9 **i** Calculate the heat energy released in this experiment. The specific heat capacity of the solution c = 4.2 J/g/°C and the mass of 1 cm^3 of mixture = 1 g. **(3)**

ii How many moles of calcium chloride dissolved in the solution? **(2)**

10 **iii** Work out the molar enthalpy change, in kJ/mol, for dissolving calcium chloride in water. **(2)**

SKILLS REASONING 9

c Another student dissolved magnesium chloride in water. She compared her result with a data book value.

Student's value = −105 kJ/mol

Data book value = −141 kJ/mol

There are no errors in the calculation of her result.

Suggest two reasons why the student's value differs from the data book value. **(2)**

(Total 13 marks)

SKILLS PROBLEM SOLVING 9

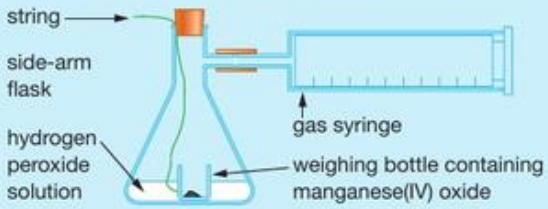
3 In an experiment to investigate the rate of decomposition of hydrogen peroxide solution in the presence of manganese(IV) oxide, a student mixed 10 cm^3 of hydrogen peroxide solution with 30 cm^3 of water and added 0.20 g of manganese(IV) oxide. She measured the volume of oxygen evolved at 60 s intervals. The results of her experiment are recorded in the table below.

Time/s	0	60	120	180	240	300
Volume/cm^3	0	30	48	57	60	60

a Write a balanced equation for the decomposition of hydrogen peroxide. **(2)**

SKILLS REASONING 6

b Explain why the manganese(IV) oxide was added in a weighing bottle rather than directly into the hydrogen peroxide solution. **(1)**



SKILLS INTERPRETATION 7

c Plot a graph of her results and draw a line of best fit. Make sure that you label the axes. **(4)**

SKILLS ANALYSIS, PROBLEM SOLVING

d Use your graph to find out the following:

5 **i** How long it took to produce 50 cm^3 of oxygen. **(1)**

ii The volume of gas produced after 100 seconds. **(1)**

6 **iii** The average rate (with unit) in the first 150 seconds. **(2)**

e Explain why the graph becomes horizontal after 240 seconds. **(2)**

f Manganese(IV) oxide acts as a catalyst for this reaction.

SKILLS CRITICAL THINKING 7

i A catalyst speeds up a chemical reaction by (choose one answer): **(1)**

A increasing the number of collisions between the reactant particles.

B providing an alternative reaction pathway with lower activation energy.

C increasing the energy of the reactant particles.

D changing the enthalpy of the reaction.

SKILLS DECISION MAKING 9

ii Describe a method you could use to show that the manganese(IV) oxide is acting as a catalyst in this reaction. **(4)**

SKILLS INTERPRETATION 7

g Suppose the experiment had been repeated using the same quantities of everything, but with the reaction flask immersed in ice.
Sketch the graph you would expect to get. Use the same grid as in **c**.
Label the new graph **G**. **(2)**

h On the same grid as in **c** and **g** sketch the graph you would expect to get if you repeated the experiment at the original temperature using 5 cm^3 of hydrogen peroxide solution, 35 cm^3 of water and 0.20 g of manganese(IV) oxide. Label this graph **H**. **(2)**

(Total 22 marks)

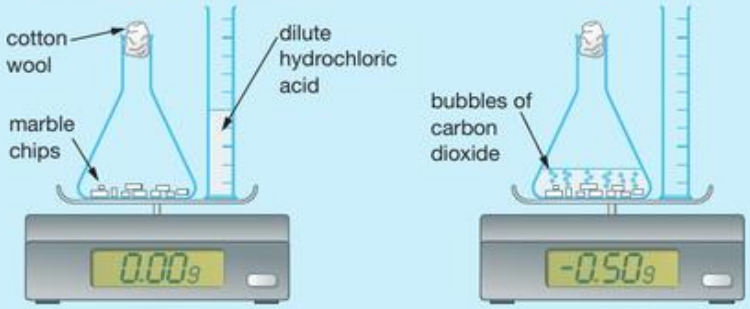
SKILLS EXECUTIVE FUNCTION 6

4 A group of students investigated the effect of changing the concentration of dilute hydrochloric acid on the rate of its reaction with marble chips (calcium carbonate).

The equation for this reaction is

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

They used the following method:



a) before reaction b) during reaction

The students recorded the time taken for the mass of the flask and contents to decrease by 0.50 g.

The experiment was then repeated using different concentrations of hydrochloric acid.

a To ensure it was a valid (fair) test, the students kept the number of marble chips constant in each experiment. Suggest two other properties of the marble chips that should be kept the same in each experiment. **(2)**

SKILLS CRITICAL THINKING 5

b What is the purpose of the cotton wool? **(1)**

c The teacher gave the students some hydrochloric acid that was labelled 100%. The table below shows the results.

Students	Mass of $CaCO_3$/g	Volume of 2.00 mol/dm^3 HCl/cm^3	Volume of water/cm^3	Concentration of HCl/%	Time to lose 0.50 g/s
1	5.00	25	0	100	105
2	5.00	20	5	80	150
3	5.00	15		60	175
4	5.00	10	15	40	272
5	5.00	5	20	20	520

SKILLS PROBLEM SOLVING 6

i The results of Student 3 are incomplete.
Calculate the volume of water the student should have used for the result to be comparable with the other four (choose one answer). **(1)**

A 5
B 10
C 15
D 20

SKILLS INTERPRETATION 7

ii Plot a graph of the results on a separate piece of graph paper, with concentration of acid on the *x*-axis and time on the *y*-axis.
Draw a line of best fit. **(4)**

SKILLS ANALYSIS, PROBLEM SOLVING 5

SKILLS REASONING

iii Use your graph in ii to find the time taken for the loss of 0.50 g of mass from the flask when the concentration of acid is 70%. **(1)**

iv One of the points on the graph is anomalous. Identify a reason for this (choose one answer). **(1)**

A The student started the stopwatch too late.

B The student stopped the stopwatch before mass loss reached 0.50 g.

C The student added too much water at the beginning of the experiment.

D The student spilt some water before adding it into the reaction mixture.

d Another group of students repeated the experiment, but this time they measured the mass loss after 1 minute.

The table below shows the results obtained by the students.

Mass of carbon dioxide given off/g	0.36	0.72	0.88	1.28	1.44	1.65
Concentration of acid /mol/dm^3	0.20	0.40	0.50	0.70	0.80	0.90

i Describe the relationship between the mass of carbon dioxide given off in 1 minute and the concentration of the acid. **(2)**

7 ii Explain this relationship in terms of particles. **(3)**

(Total 15 marks)

5

During the manufacture of nitric acid from ammonia, the ammonia is oxidised to nitrogen monoxide, NO, by oxygen in the air.

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$$

The ammonia is mixed with air and passed through a stack of large circular gauzes made of platinum–rhodium alloy at red heat (about 900 °C). The platinum–rhodium gauzes act as a catalyst for the reaction.

a Gas particles have to collide before they can react. Use the collision theory to help you to answer the following questions.

i Because the gases are in contact with the catalyst for only a very short time, it is important that the reaction happens as quickly as possible. Explain why increasing the temperature to 900 °C makes the reaction very fast. **(3)**

ii Explain what will happen to the reaction rate if the pressure is increased. **(2)**

iii Explain why the platinum–rhodium alloy is used as gauzes rather than as pellets. **(2)**

b Platinum and rhodium are extremely expensive metals. Explain why the manufacturer can justify their initial cost. **(2)**

(Total 9 marks)

6

At temperatures above 150 °C, brown nitrogen dioxide gas dissociates (splits up reversibly) into colourless nitrogen monoxide and oxygen:

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g) \qquad \Delta H = +114 \text{ kJ/mol}$$

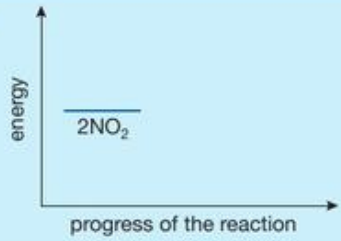
a Write down the meaning of the symbols $\rightleftharpoons$ and ΔH. **(2)**

b What does the positive sign of ΔH indicate about the reaction? **(1)**

CHEMISTRY ONLY

SKILLS INTERPRETATION (9)

c Complete the reaction profile diagram for this reaction.



Label the diagram clearly with the activation energy and the ΔH of the reaction. **(3)**

SKILLS REASONING (8)

d Predict what would happen to the position of equilibrium and the colour of the mixture if you:

i increase the temperature **(3)**

ii increase the pressure **(3)**

e Reactions can be accelerated using a catalyst. Write down and explain the effect of a catalyst on the position of equilibrium in this reaction. **(2)**

(Total 14 marks)

(6) **7** Nitrogen and hydrogen are used in the manufacture of ammonia (NH_3).

The reaction is reversible and can reach a state of dynamic equilibrium:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

SKILLS CRITICAL THINKING

a What are the characteristics of a reaction at dynamic equilibrium (choose two answers)? **(2)**

A The rate of the forward reaction is the same as the rate of the reverse reaction.

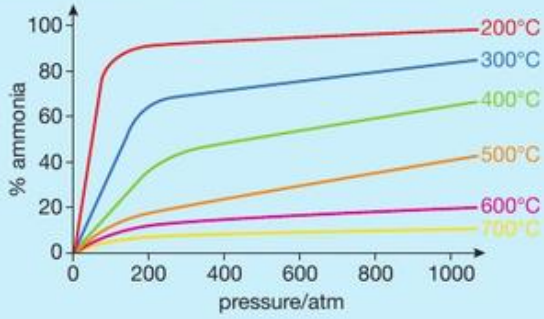
B The concentrations of the reactants and the products are always equal.

C The position of equilibrium can be shifted by adding a catalyst.

D The concentrations of the reactants and the products are constant.

SKILLS REASONING

b The graph below shows how the percentage of NH_3 in the equilibrium mixture varies with temperature and pressure (in atmospheres (atm)).



(8) i Increasing pressure increases the percentage of NH_3 at equilibrium. Explain why this is the case. **(3)**

(5) ii Describe the relationship between temperature and the percentage of NH_3 at equilibrium. **(1)**

(8) iii Using your answer from ii, comment on the sign of enthalpy change (ΔH) for the forward reaction. **(2)**

iv Predict and explain what happens to the rate of the reaction when the temperature is increased. **(3)**

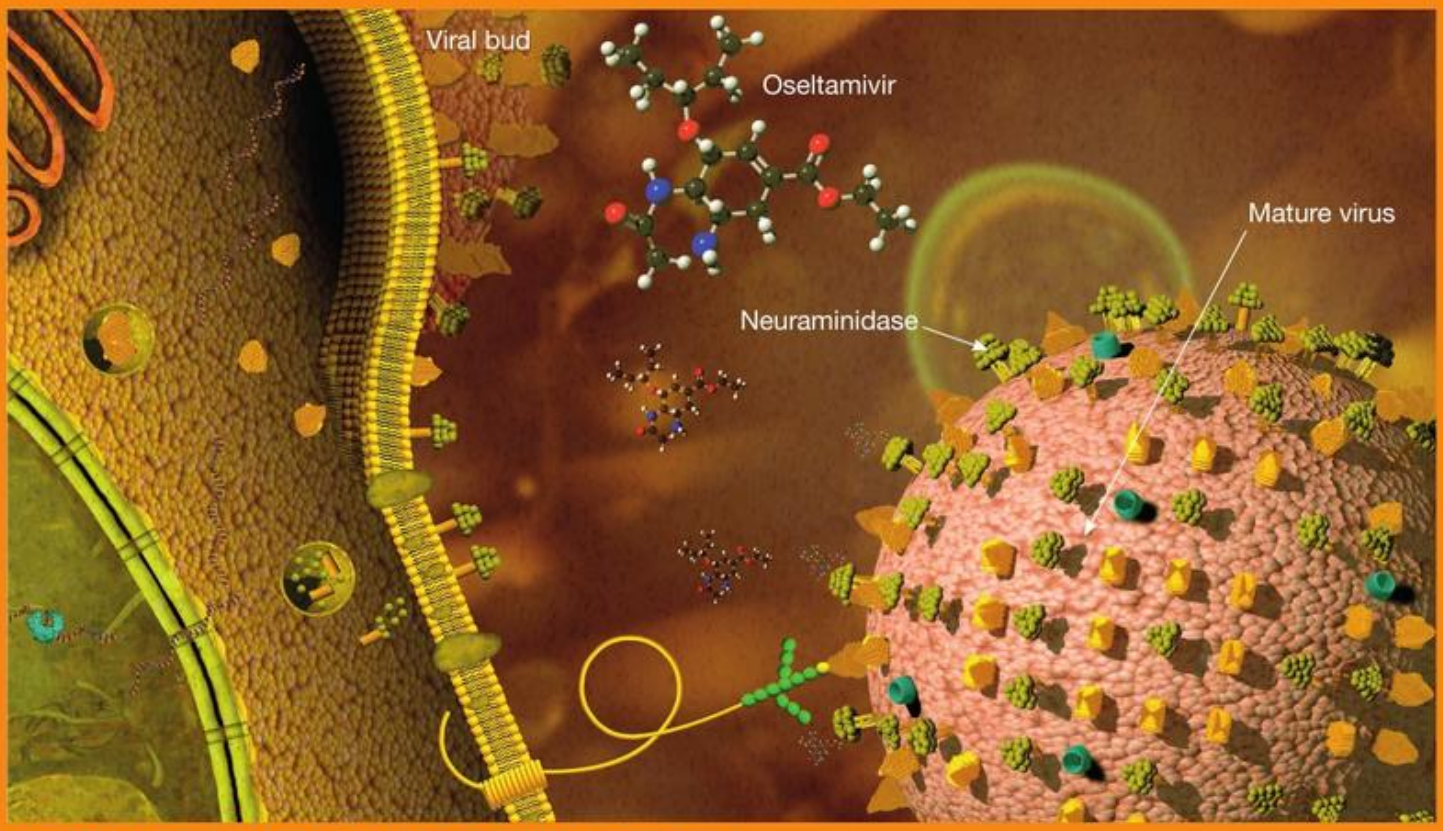
(Total 11 marks)

END OF CHEMISTRY ONLY

INTRODUCTION TO ORGANIC CHEMISTRY 255 | CRUDE OIL 268 | ALKANES 277 | ALKENES 282 | ALCOHOLS 287
CARBOXYLIC ACIDS 293 | ESTERS 297 | SYNTHETIC POLYMERS 302

UNIT 4
ORGANIC CHEMISTRY

Organic chemistry is the study of the compounds of carbon (there are some inorganic carbon compounds as well). There are more organic compounds than all the other compounds put together. These compounds include naturally occurring ones that are found in our bodies, for example proteins, DNA and fats. There are also artificially made compounds such as plastics, dyes and drugs. The artificial compounds are derived from chemicals obtained from crude oil. When all the crude oil has been used we will have to find new sources for more than just fuel for cars and making electricity. Organic chemists are involved in the synthesis of a huge variety of new compounds, including drugs and medicines used to treat diseases such as cancer, AIDS, influenza and asthma. The synthesis of these drugs can involve a large number of steps and requires the knowledge and understanding of a great number of organic chemistry reactions. It can take many years and millions of dollars to develop a new drug.



▲ Figure 22.1 Oseltamivir (Tamiflu®) is a drug that targets the influenza virus.